Harvard John A. Paulson School of Engineering and Applied Sciences

Bayesian Inverse Transition Learning for Offline Settings [1]

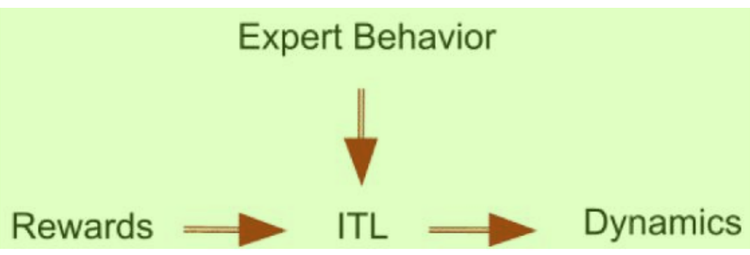
Leo Benac, Sonali Parbhoo, Finale Doshi-Velez [2]

Summary [3]

- Learning a model is hard under low-coverage data [4]
- Having a model help be more data efficient
- Inverse Transition Learning infers the dynamics using near-optimal demonstrations
- We create a set of constraints to estimate in a gradient-free way an empirical posterior distribution over the transition dynamics T, where each sample has near-optimal guarantees
- Show that planned policies are highly performant and exhibit lower variance across different batches of datas compared to MLE estimates
- Explain how quantifying the uncertainty of the dynamics help planning more informative policies than the expert policy and the one obtained from MLE estimates

What is Inverse Transition Learning? [5]

Inverse Transition Learning



[6]

Infer the dynamics T given near-optimal demonstrations $\mathcal{D}$, the reward function R and a near-optimal expert policy π_ϵ, where π_ϵ is assumed to only take actions that are ϵ-close to the best unkown action a^*. [7]

Why is it hard? [8]

- Number of parameters grows with the dimensions of state and action spaces [9]
- Trade-off between a model that is expressive enough and one that is small enough to learn
- Lack of coverage in offline inverse settings

Why do we care? [10]

- Data efficiency [11]
- Simulation of new data
- Off-Policy Evaluation
- Counterfactual Reasoning

Baselines [12]

- T^{MLE} Maximum Likelihood Estimate obtained from the expert demonstrations $\mathcal{D}$ [13]
- $P(T(s'|s,a)|\mathcal{D}) = Dir(N_{s,a} + \mathbf{1}|s,a)$ Probabilistic model over T without making any optimality assumption. Where $\mathbf{1}$ is a vector of 1's of dimension $|S|$ and $N_{s,a}$ is the number of transitions from state s and action a in the batch data $\mathcal{D}$

Constraints [14]

Using the reward R and near-optimal expert policy π_ϵ we create a set of constraints with respect to the dynamics T to impose the values of the actions of the expert to be superior to the ones of unseen actions. [15]

Method [16]

- Sample from $P(T|\mathcal{D})$ and only keep samples that satisfies the constraints. [17]
- Create empirical posterior $P(T|\mathcal{D}, \pi_\epsilon)$ with accepted samples. This will be the estimate of the posterior on T given that the demonstrations from the expert are near-optimal

Metric [18]

$$Q^*_{metric}(\hat{\pi}) = \sum_{s \in S} \left(Q^*(s, a^*) - \sum_{a \in A} \hat{\pi}(a|s) Q^*(s,a) \right)$$ [19]

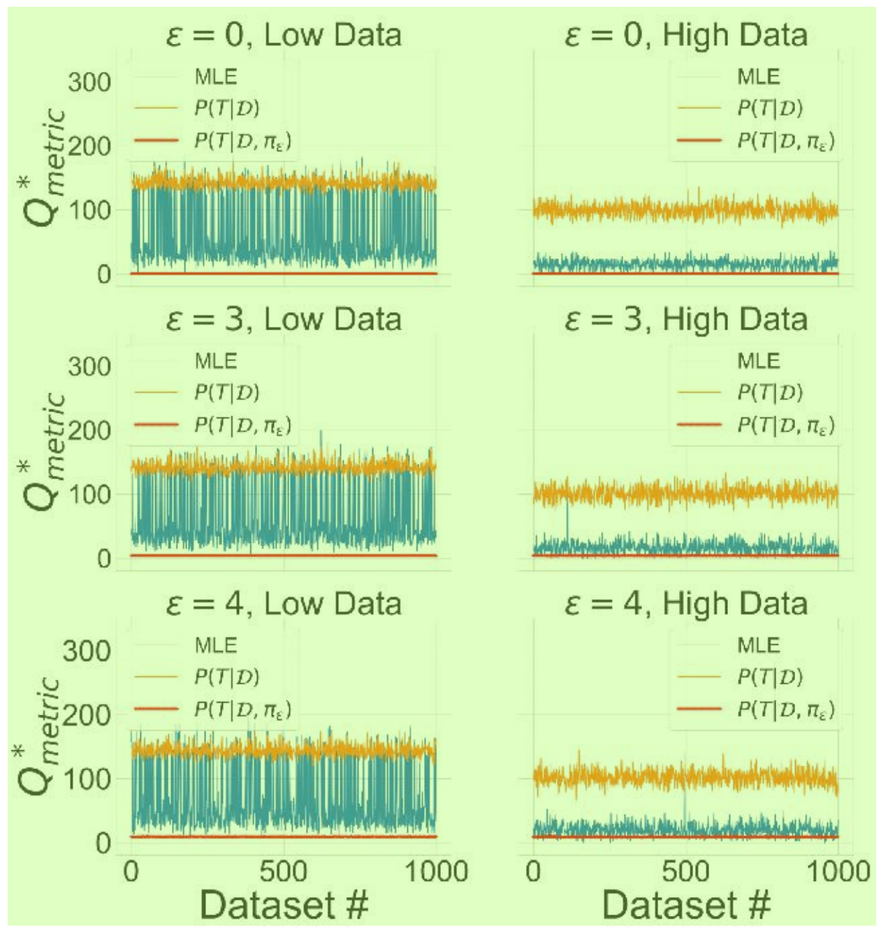
Results [20]

- **MLE** [21]

$$\hat{T} = T^{MLE} \xrightarrow[\text{Value Iteration}]{} (\pi^{MLE}, Q^{MLE}) = (\hat{\pi}, \widehat{Q}) \xrightarrow[\text{Compute Results}]{}$$ [22]

- $P(T|\mathcal{D}, \pi_\epsilon)$ **and** $P(T|\mathcal{D})$ [23]

$$\{\hat{T}^{(i)}\}_{i=1}^{1000} \sim \text{Posterior} \xrightarrow[\text{Value Iteration}]{} \{(\hat{\pi}^{(i)}, \widehat{Q}^{(i)})\}_{i=1}^{1000} \xrightarrow[\text{Empirical Mean}]{} \{(\hat{\pi}, \widehat{Q})\} \xrightarrow[\text{Compute Results}]{}$$ [24]



[25]

- $P(T|\mathcal{D}, \pi_\epsilon)$ outerperforms both baselines by being more accurate and avoiding low value actions [26]
- $P(T|\mathcal{D}, \pi_\epsilon)$ exhibits significantly less variance over different datasets
- $P(T|\mathcal{D}, \pi_\epsilon)$ allows quantifying uncertainty on corresponding policies

Acknowledgements [27]

This material is based upon work supported by the National Science Foundation under Grant No. IIS-2007076. Any opinions, findings, and conclusions or recommendations expressed in this material are those of the author(s) and do not necessarily reflect the views of the National Science Foundation. [28]